A Sharp M-Matrix Threshold for a Two-Dimensional Oblique Skorokhod Map

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Abstract

For oblique regulation in the quadrant with the stated two-axis reflection matrix R, we prove the *fixed point and sharp M-matrix chamber*: every càdlàg input has one and only one regulated path precisely when $\rho\sigma < 1$. We add *weighted stability, Picard, and time change*: the natural norm has contraction factor $\sqrt{\rho\sigma}$, explicit solution stability, monotone geometric iteration, causality and reparameterization covariance.

1 Convention and the sharp existence theorem

Fix $T < \infty$, $\rho, \sigma \geq 0$, and

$$R = \begin{pmatrix} 1 & -\rho \\ -\sigma & 1 \end{pmatrix}.$$

Let $x = (x_1, x_2) \in D([0,T], \mathbb{R}^2)$ with $x(0) \geq 0$ coordinatewise. A solution consists of càdlàg z, y such that

$$z = x + Ry \geq 0, \qquad y(0) = 0, \qquad y_i \text{ is nondecreasing}, \qquad \int_{[0,t]} z_i(s)\, dy_i(s) = 0. \tag{1}$$

The integrand is the *post-jump* state. Thus a regulator jump is supported on an axis after the jump, including simultaneous corner jumps.

For scalar càdlàg f, write

$$(\mathcal{L}f)(t) = \sup_{0 \leq s \leq t} [-f(s)]_+.$$

Lemma 1 (One-sided regulator)**.** *If $f(0) \geq 0$, the unique nondecreasing càdlàg r, with $r(0) = 0$, for which $g = f + r \geq 0$ and $\int g\, dr = 0$ is $r = \mathcal{L}f$. Moreover,*

$$\|\mathcal{L}f - \mathcal{L}h\|_\infty \leq \|f - h\|_\infty. \tag{2}$$

Proof. The running supremum is the least nondecreasing correction making $f + r$ nonnegative. It increases only at a new negative running minimum, where the post-jump corrected state is zero. Conversely, feasibility forces every regulator to dominate $\mathcal{L}f$. A first strict excess would be an increase with positive corrected state, contradicting complementarity. Finally, positive part and running supremum are one-Lipschitz, proving (2). ☐

Theorem 2 (Sharp all-input threshold)**.** *Problem (1) has exactly one solution for every admissible input and every finite horizon if and only if*

$$\rho\sigma < 1. \tag{3}$$

Inside this chamber, its regulator is the unique solution of

$$y_1 = \mathcal{L}(x_1 - \rho y_2), \qquad y_2 = \mathcal{L}(x_2 - \sigma y_1). \tag{4}$$

Proof. With the other coordinate held fixed, each line of (1) is the scalar problem in Lemma 1; hence (1) and (4) are equivalent. Conversely, at time zero the fixed-point equations and $x(0) \geq 0$ imply $y_1(0) \leq \rho y_2(0)$ and $y_2(0) \leq \sigma y_1(0)$; since $\rho\sigma < 1$, both values vanish. Thus the fixed point has the required initial condition.

First suppose $\rho, \sigma > 0$, put $w_1 = \sqrt{\rho}$, $w_2 = \sqrt{\sigma}$, $q = \sqrt{\rho\sigma}$, and define

$$\|u\|_w = \max_{i=1,2} \frac{\|u_i\|_\infty}{w_i}.$$

For the right side Φ_x of (4), inequality (2) gives

$$\|\Phi_x(u) - \Phi_x(v)\|_w \le q\|u - v\|_w.$$

Thus (3) makes Φ_x a contraction on the complete space of bounded càdlàg pairs. Its unique fixed point is càdlàg and solves the problem. The constant q is the best uniform coefficient: each weighted cross-coordinate estimate can be attained by a constant difference after a suitable downward input shift.

If $\rho = 0$, solve successively $y_1 = \mathcal{L}x_1$ and $y_2 = \mathcal{L}(x_2 - \sigma y_1)$; if $\sigma = 0$, reverse the order. This also covers $\rho = \sigma = 0$ and proves sufficiency.

For necessity, let $\rho\sigma = 1$. Then $R(\rho, 1)^{\mathsf{T}} = 0$, so zero input has the distinct solutions

$$y = (\rho h, h), \qquad z = 0, \tag{5}$$

for every nondecreasing càdlàg h with $h(0) = 0$. More strongly, whenever $\rho\sigma \ge 1$, an input jumping from zero to $(-1, -1)$ cannot be feasible: the post-jump values would obey

$$y_1 \ge 1 + \rho y_2, \qquad y_2 \ge 1 + \sigma y_1,$$

and hence $(1 - \rho\sigma)y_1 \ge 1 + \rho$, an impossibility. Therefore no point on or beyond the wall is well posed for every input. □

2 Quantitative stability and path structure

Theorem 3 (Stability and constructive convergence)**.** *Assume $\rho, \sigma > 0$ and $q < 1$. If inputs x, x' have solutions $(z, y), (z', y')$, then*

$$\|y - y'\|_w \le \frac{\|x - x'\|_w}{1 - q}, \qquad \|z - z'\|_w \le \frac{2\|x - x'\|_w}{1 - q}. \tag{6}$$

Starting with $y^{(0)} = 0$ and setting $y^{(n+1)} = \Phi_x(y^{(n)})$ gives a coordinatewise increasing sequence and

$$\|y^{(n+1)} - y^{(n)}\|_w \le q^n \|y^{(1)}\|_w, \tag{7}$$

$$\|y - y^{(n)}\|_w \le \frac{q^n}{1 - q}\|y^{(1)}\|_w. \tag{8}$$

The solution map is causal, preserves continuity, and commutes with continuous nondecreasing onto time changes that preserve the endpoints.

Proof. For two inputs, the scalar estimate gives

$$\|\Phi_x(u) - \Phi_{x'}(v)\|_w \le \|x - x'\|_w + q\|u - v\|_w.$$

Evaluate it at the two fixed points for the first inequality in (6). Each normalized row of R has weighted absolute sum $1 + q$, whence

$$\|R(y - y')\|_w \le (1 + q)\|y - y'\|_w.$$

Using $z - z' = x - x' + R(y - y')$ yields the second inequality. The map Φ_x is order preserving and $0 \le \Phi_x(0)$, so Picard iterates increase. Contraction gives (7), and summing its geometric tail gives (8).

Running suprema use only the past, so uniqueness gives causality. They map continuous functions to continuous functions; uniform Picard convergence then preserves continuity. If $\lambda : [0, T'] \to [0, T]$ is continuous, nondecreasing, onto, and endpoint preserving, its initial images satisfy $\lambda([0, t]) = [0, \lambda(t)]$. Consequently

$$\mathcal{L}(f \circ \lambda) = (\mathcal{L}f) \circ \lambda.$$

Compose (4) with λ and use uniqueness. Flat pieces of λ simply pause both state and regulator. □

Remark 4. *When one weight vanishes, the weighted norm is not defined. The triangular formulas in Theorem 2 are the correct boundary statement; no singular limiting norm is smuggled into that face.*

Source lineage

This source-local reconstruction makes no literature-priority claim.

References

[1] J. M. Harrison and M. I. Reiman, "Reflected Brownian Motion on an Orthant," *Annals of Probability* 9 (1981), 302–308. DOI: 10.1214/aop/1176994471.

[2] P. Dupuis and H. Ishii, "On Lipschitz Continuity of the Solution Mapping to the Skorokhod Problem, with Applications," *Stochastics and Stochastics Reports* 35 (1991), 31–62. DOI: 10.1080/17442509108833688.